A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

25_26J_009

Project Proposal Report

Wijekoon H.M.R.K.

B.Sc. (Hons) in Information Technology Specializing in Computer Systems & Network Engineering

Department of Computer Systems Engineering

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate dissertation under my supervision.

Signature of the Supervisor: ..... Date 24/08/25

Signature of the Co-Supervisor: ..... Date 24/08/2025

Acknowledgement

Working on this final-year research project has been an immensely enriching academic experience for me. Through the process of developing my research proposal, I have gained valuable insights into the research process, including how to formulate research questions, design suitable methodologies, and anticipate potential challenges.

I would like to express my heartfelt gratitude to my supervisor, Ms. Shashika Lokuliyana, for her invaluable guidance, feedback, and support throughout this project. Her constant input and insightful suggestions are greatly appreciated. I would also like to express my gratitude to my co-supervisor, Mr. Anuradha Jayakody, for his unwavering guidance, feedback, and support throughout the project. I owe a debt of gratitude to all the lecturers at SLIIT as well.

I look forward to continuing to develop my research skills and contribute meaningful insights through this project.

Abstract

The Internet of Things (IoT) has revolutionized modern infrastructure through interconnected devices, but the inherent vulnerabilities in the Routing Protocol for Low-Power and Lossy Networks (RPL) pose significant security challenges. This research addresses the critical need for trust-aware routing mechanisms in RPL-based IoT networks by developing an enhanced objective function that integrates trustworthiness and security factors into routing decisions.

This research proposes the design and implementation of a trust and security-aware RPL routing mechanism as part of a lightweight trust management framework. The solution introduces an enhanced objective function that integrates trustworthiness and security factors into parent selection and routing decisions. In addition, a dynamic parent selection algorithm will be developed to ensure adaptability in dynamic network conditions, while secure route broadcasting will be implemented to strengthen resilience against routing attacks. The approach will be evaluated through simulations using Contiki-NG and its Cooja simulator, which provide realistic modeling of IoT networks and attack scenarios. By combining trust computation, attack detection, and secure routing decisions, the proposed framework aims to overcome the limitations of traditional RPL while remaining efficient for resource-constrained IoT devices.

Expected outcomes include improved trustworthiness, network security, reduced attack success rates (mitigation, isolation, and recovery), and maintained routing efficiency while preserving the lightweight nature required for resource-constrained IoT devices. This work contributes to the broader lightweight trust management framework for IoT security by providing a robust, distributed routing solution that enhances network resilience against threats.

Keywords – Internet of Things (IoT), RPL, Trust Management, Secure Routing, Contiki-NG

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1.0 INTRODUCTION

The Internet of Things (IoT) has become one of the most influential technologies in recent years, changing the way people, devices, and systems interact with each other. It connects everyday objects such as sensors, actuators, and gateways to form large networks that can collect and exchange data. These networks are now widely used in different areas, including smart homes, healthcare monitoring, industrial automation, disaster management, and smart cities. Analysts predict that by 2025, the number of IoT devices worldwide will exceed 75 billion, highlighting the growing importance of this field [7], [9], [24].

For these networks to function reliably, they require efficient and secure communication. However, most IoT devices are highly resource-constrained, meaning they have limited processing power, memory, and energy. To support communication in such environments, the Internet Engineering Task Force (IETF) introduced the Routing Protocol for Low-Power and Lossy Networks (RPL) [21]. RPL is now the standard routing protocol in IoT environments, as it is lightweight and capable of building efficient paths for data transmission across large numbers of devices [21]–[23].

While RPL has proven to be effective for routing, it was not originally designed with strong security features in mind [2], [3], [24]. This has left IoT systems vulnerable to several types of routing attacks, such as rank attacks, sinkhole attacks, and selective forwarding, which can reduce network performance or even compromise data integrity [2], [3], [19]. These risks are especially serious in critical applications like emergency response or healthcare, where any loss or delay of information can have serious consequences [1], [24].

Because of these challenges, there is a clear need to improve the security of RPL-based IoT networks without making them too complex or resource-intensive. This research focuses on exploring how trust and security can be integrated into RPL at the routing layer to create more resilient and dependable IoT systems [1]–[3].

1.1 Background

The Internet of Things (IoT) is built on the idea of connecting a wide range of devices so they can sense, collect, and share data. These devices often work in environments where energy, memory, and processing power are limited. Because of these constraints, IoT networks rely on lightweight communication protocols to ensure data can travel reliably from devices to central systems.

The Routing Protocol for Low-Power and Lossy Networks (RPL) was developed by the Internet Engineering Task Force (IETF) as a standardized solution for such scenarios [21]. RPL organizes devices into a tree-like structure, called a Destination-Oriented Directed Acyclic Graph (DODAG), which makes it easier to transmit data to a central root node [21]. It uses objective functions (OF) to select the best communication paths, considering factors such as hop count, link quality, or energy use [22], [23].

Although RPL is widely used and efficient, it was designed mainly with performance and energy savings in mind, rather than security [2], [3]. As IoT deployments have grown in scale and importance, this design limitation has become a major concern. Attackers can exploit weaknesses in the routing process, causing issues like packet loss, false routing information, or complete network disruptions [2], [3], [19]. These vulnerabilities are especially critical in domains like disaster monitoring, healthcare, and industrial control, where reliability and trustworthiness of data are essential [24].

This background highlights the importance of strengthening RPL against security threats. To do this effectively, new approaches must combine efficiency with robust protection, so that IoT devices remain functional within their strict resource limits while ensuring secure communication [1]–[3].

1.2 Literature Review

RPL is widely recognized as the standard routing protocol for IoT deployments due to its lightweight nature and adaptability to constrained devices [21]–[23]. However, its original design prioritizes energy efficiency and routing reliability rather than strong security, making it vulnerable to rank manipulation, sinkhole attacks, selective forwarding, Sybil, and blackhole attacks [2], [3], [19], [24]. These vulnerabilities have motivated researchers to propose trust and security-aware routing mechanisms.

Trust-based enhancements to RPL, Recent work integrates trust scores into parent selection, often combining residual energy, historical behaviour, and communication quality. Examples include enhanced objective functions and trust-aware protocols that improve resilience but may increase complexity or convergence time [10]–[13], [17]. CTrust-RPL, for instance, targets secure routing with a control-layer trust mechanism, while multi-dimensional models (e.g., SMTrust) incorporate energy and reputation signals; nevertheless, scalability and overhead remain concerns in large IoT deployments [6], [12], [13].

Attack-focused defenses**,** Detection methods tailored to RPL routing attacks include DETONAR for routing-attack detection and targeted sinkhole-detection strategies for 6LoWPAN/IoT [5], [18]. Surveys also catalogue threats and countermeasures at protocol level, offering design guidance for defensive mechanisms [2], [19], [24].

Learning-based detection and datasets**,** Network-based anomaly detection using deep autoencoders has been applied to IoT botnet traffic (N-BaIoT) [15], while broader intrusion-detection surveys summarize techniques and commonly used datasets such as UNSW-NB15 for evaluating detection performance [14].

Synthesis, Prior work underscores two persistent issues: (i) solutions often trade security against performance (or vice-versa), which is problematic for mission-critical IoT; and (ii) parent selection can be rigid, limiting adaptability under dynamic conditions, which affects stability and end-to-end

reliability [1], [3], [17]. These gaps motivate a lightweight, balanced approach that integrates trust and security into routing while preserving core performance metrics [10], [17], [22], [23].

1.3 Research Gap

Although RPL has been widely adopted as the standard routing protocol for Low-Power and Lossy Networks (LLNs), its limited security features continue to expose IoT systems to significant risks [2], [3]. Several trust and security-aware extensions of RPL have been proposed, including SecTrust-RPL, SMTrust, CTrust-RPL, MRTS, and Trust-RPL [5]–[8], [11]. While these models highlight the potential of integrating trust into routing, they also present notable shortcomings. Many of them prioritize either security or performance, resulting in trade-offs that are unsuitable for mission-critical IoT applications. In addition, most rely on static or inflexible parent selection strategies, which reduces adaptability in dynamic IoT environments and often leads to unstable routing paths [6], [7].

Moreover, a number of existing mechanisms introduce complex trust computation models, machine learning algorithms, or blockchain-based systems [12], [13]. While these approaches may enhance security, they are resource-intensive and often impractical for highly constrained IoT devices. This reflects a gap in existing research, the lack of a lightweight and balanced solution that integrates both trust and security into RPL while maintaining routing performance.

1.4 Research Problem

Existing trust and security-aware RPL models either compromise performance for stronger security or remain too computationally demanding for resource-constrained IoT devices. In addition, most fail to provide dynamic adaptability in parent selection, resulting in unstable routing structures in changing environments. Therefore, there is a need for a practical and efficient mechanism that integrates trust and security into RPL without sacrificing essential performance metrics.

1.5 Research Questions

- How can trust and security factors be effectively integrated into the RPL objective function without introducing significant overhead for IoT devices?
- Can a dynamic parent selection mechanism improve the stability and resilience of RPL under routing attacks?
- What impact does the proposed trust and security-aware RPL mechanism have on packet delivery ratio, latency, and energy efficiency compared to the standard RPL protocol?

2.0 OBJECTIVES

2.1 Main Objective

To design and implement a trust and security-aware RPL routing mechanism that enhances resilience against routing attacks while maintaining essential performance metrics in resource-constrained IoT environments.

2.2 Sub Objectives

My component focuses on developing a trust and security-aware RPL routing mechanism within the broader lightweight trust management framework:

- To modify the RPL objective function by incorporating trustworthiness and security factors into routing decisions.

- To design and implement a dynamic parent selection algorithm that adapts to varying trust and security conditions in the network.

- To enable the broadcasting of secure route path decisions based on the proposed trust-aware objective function.

- To evaluate the proposed mechanism against standard RPL using performance metrics such as packet delivery ratio, latency, energy efficiency, and resilience to routing attacks.

3.0 METHODOLOGY

3.1 System diagram

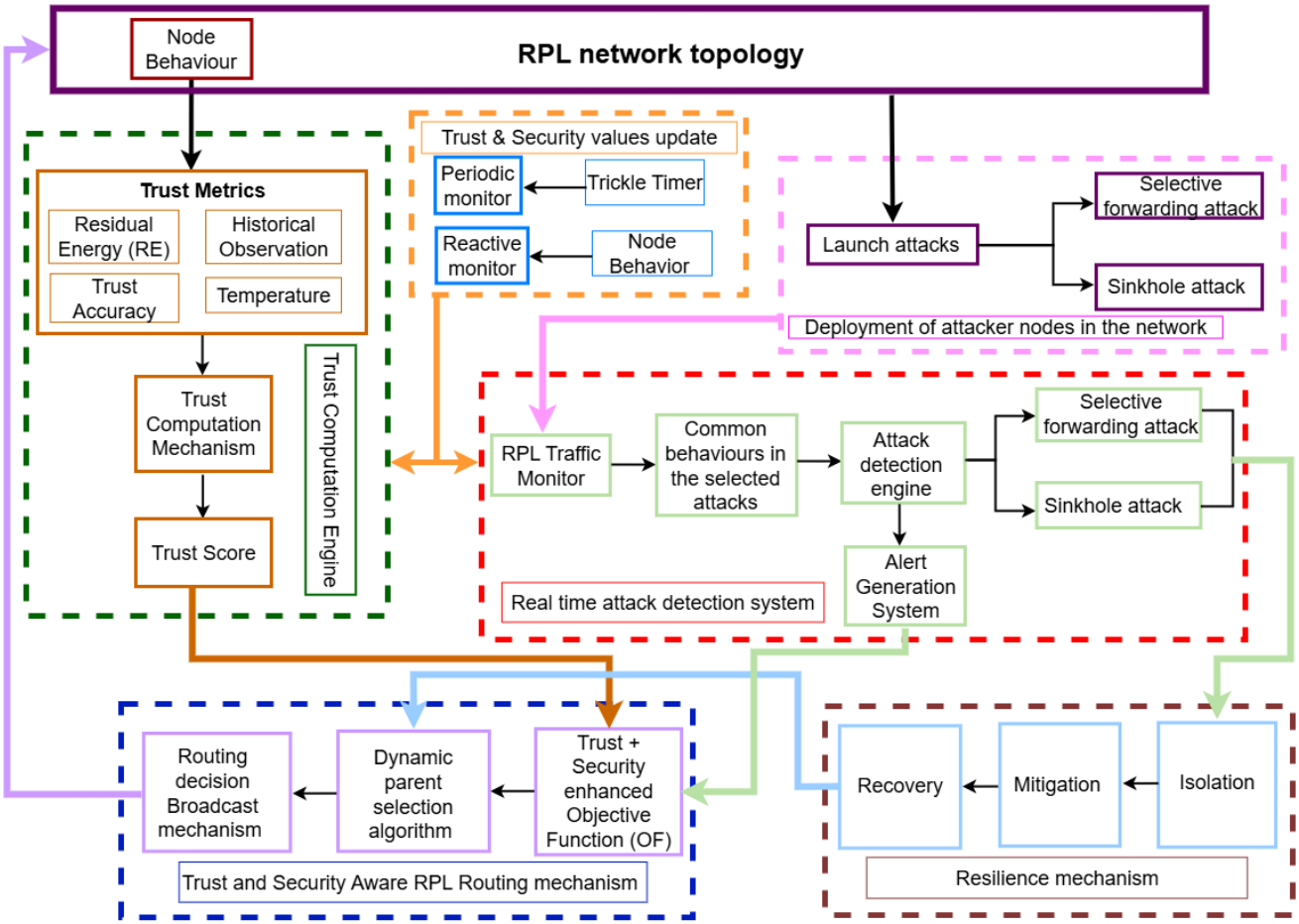

Figure 1: Lightweight Trust Management Framework for RPL-based IoT Networks

Figure 1 will present the proposed RPL network topology and the four main modules of the framework, the Trust Computation Engine, the RPL Traffic Monitor and Attack Detection Engine, the Trust & Security–Aware RPL Routing mechanism, and the Resilience Mechanism (isolation, mitigation, recovery). The data flow will begin with node behavior observation, each sensor node will monitor packet-forwarding behavior, link quality, residual energy and local indicators such as temperature. These raw observations will be input to the Trust Computation Engine, which

periodically and reactively computes a compact trust score for each neighbor using lightweight aggregation. The trust score output will then be pushed to the routing module. Simultaneously, the RPL Traffic Monitor will analyze control and data-plane flows to extract behavioral signatures, which will be fed into the Attack Detection Engine. This engine will raise alerts when suspicious patterns, such as selective forwarding or sinkhole behaviors, are detected. The routing module will then blend the trust scores, security alerts, and classical RPL metrics (ETX, hop count) into a composite "enhanced objective function" to select or re-rank parents. When a parent change is triggered, the routing module will issue secure route broadcasts so that neighboring nodes can update their local views. If the Attack Detection Engine raises a high-severity alert, the resilience module will isolate the node and trigger mitigation and recovery procedures. This layered structure will ensure that routing decisions remain adaptive, resilient, and lightweight, meeting the needs of resource-constrained IoT devices [1]–[3].

3.2 My Research Component Methodology

The core of this work will be the design and deployment of a modified RPL objective function and an adaptive parent selection policy that together will ensure secure routing decisions with minimal overhead. The modified objective function will be formulated so that a node's composite rank is a bounded combination of a traditional metric (such as ETX or base rank), a normalized trust term from the Trust Engine, and a security factor derived from detection alerts.

The trust term will be computed as a short-decay score that captures recent forwarding behavior and stability, together with one or two device health indicators (e.g., residual energy). The security factor will be event-driven and conservative, raising a parent's effective cost after a confirmed alert. Occasional false alarms will not immediately disrupt topology stability. To avoid oscillation, the parent selection policy will incorporate hysteresis windows and guard timers: small trust drops will not trigger immediate parent switching, whereas confirmed malicious behavior or repeated failures will force expedited re-selection.

All design decisions (including which metrics to combine, normalization ranges, and threshold values) will be parameterized so that they can be tuned during experiments. The implementation will modify the RPL module in Contiki-NG by extending neighbor tables to include trust/security fields and by adding the composite ranking step to the parent-ranking routine. The modified DIO/DAO behavior for broadcasting trust-aware routing decisions will be designed to remain backward-compatible, and lightweight message authentication will be integrated to protect against manipulation of trust-related fields without imposing heavy cryptographic costs.

3.3 Tools, Software, and Environment

The implementation will be carried out using Contiki-NG, an open-source operating system widely used for IoT simulations due to its accurate modeling of RPL behavior and support for low-power lossy networks. Its Cooja simulator provides node-level energy models, traffic visualization, and customizable attack scenarios, making it highly suitable for this research [24][8]. Experiments will be conducted on personal PCs with at least 16 GB RAM and 512 SSD, allowing multiple simulation runs in parallel. Development will be in C for the protocol stack modifications and Python for simulation orchestration, data collection, and statistical analysis. Code versioning will be managed using GitHub, while Microsoft Teams and WhatsApp will support group collaboration.

3.4 Data needs and collection strategy

All primary data will be produced by the Cooja/Contiki simulations: per-packet traces, per-node energy counters (simulated), parent selection logs, DIO/DAO counts, trust values over time and attack alert traces. To create realistic attack patterns and validate detection thresholds, public datasets such as IoT-23 and UNSW-NB15 will be used as references for traffic patterns and malicious behavior synthesis, while Intel Lab and SmartSantander traces are used to select sensible traffic models and sensor sampling intervals [14][9]. Each scenario will be repeated multiple times (we will

use at least 30 runs per configuration) to provide statistical confidence; logs are exported as CSV files and processed in Python to compute means, standard deviations and confidence intervals.

3.5 Experimental design and evaluation criteria

Experiments will compare the proposed mechanism against standard RPL (OF0 and MRHOF) under four primary scenarios: normal (no attack), with attack (sinkhole or selective forwarding), with Trust issues, and mixed attacks + Trust. Network sizes will be scaled (for example, 20, 50 and 100 nodes) to study scalability. Metrics include packet delivery ratio, end-to-end latency, control overhead (DIO/DAO counts), parent-switch frequency (stability), estimated energy overhead, detection rate and false positive rate, and time-to-mitigation. Success will be measured by the system's ability to maintain delivery and latency performance within acceptable deviation from baseline RPL, while significantly improving resilience against attacks. Statistical tests will be conducted to confirm the validity of results [10] [11] [14] [17].

3.6 Tasks, sub-tasks and milestones

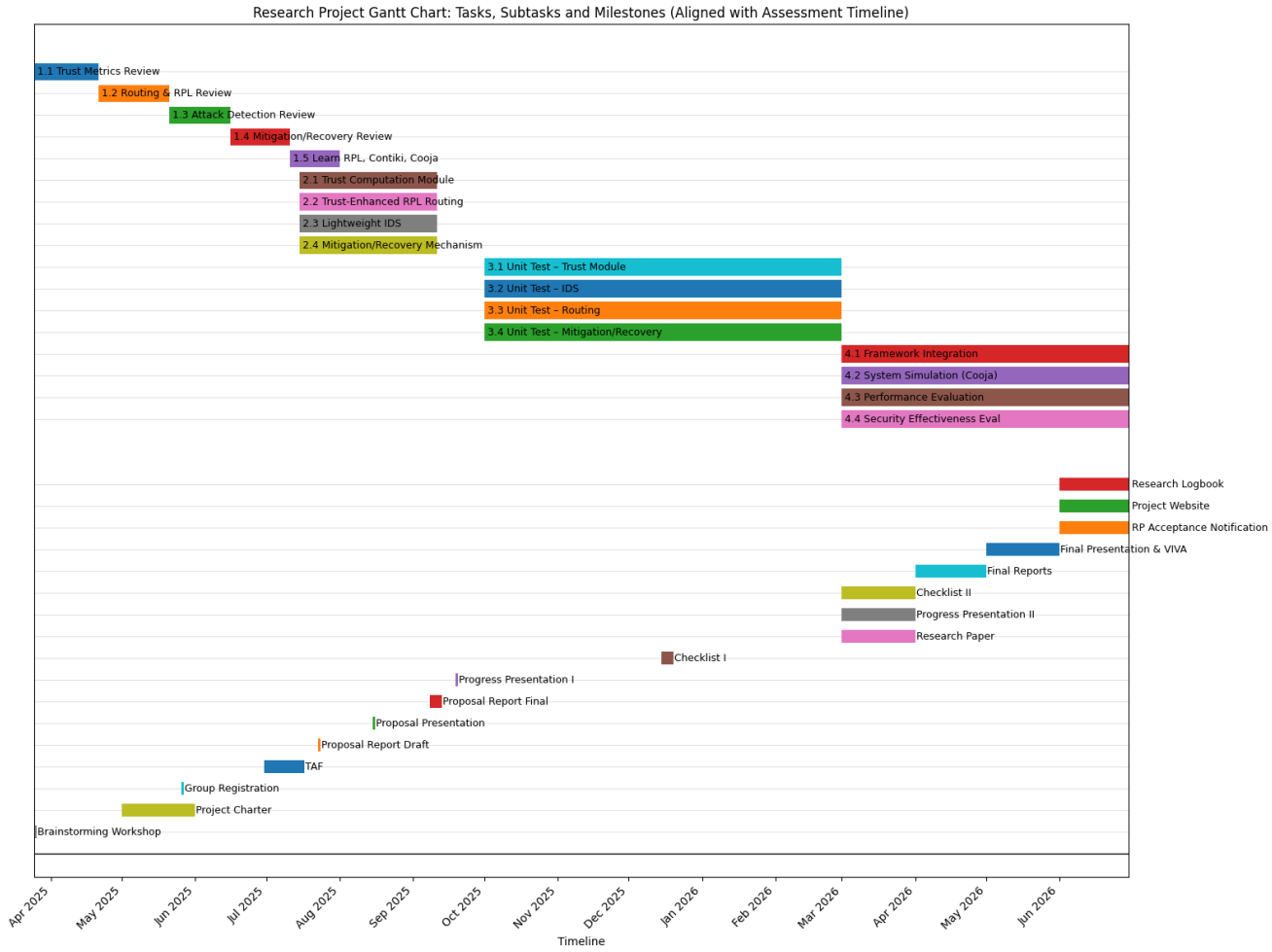

Figure 2: Overall Research Timeline (Gantt Chart)

The first phase focuses on literature review and requirements definition, synthesizing existing work on RPL security, trust metrics, and intrusion detection systems to establish the foundation for design. This phase also defines evaluation criteria such as packet delivery ratio, energy efficiency, and attack resilience.

The second phase involves algorithm and model design. Here, the mathematical formulation of the composite objective function is developed, and the parent selection algorithm is defined with stability

mechanisms such as guard timers. The design phase also specifies secure broadcast formats and lightweight authentication to protect against manipulation of trust-related control messages [7].

The third phase is the implementation, where the enhanced objective function and parent selection policies are integrated into Contiki-NG. The implementation also includes secure broadcast handling, data logging mechanisms for simulation, and APIs to integrate the trust and detection modules.

The fourth phase is framework integration, where modules from group members such as the Trust Engine and Attack Detection System—are combined into a single system. This requires defining message formats, synchronization of update intervals, and collaborative debugging.

The final phase is evaluation, where the complete system is tested under multiple IoT scenarios with varying network sizes and attack intensities. Metrics such as delivery ratio, latency, control overhead, and resilience will be measured, with statistical analysis (t-tests or non-parametric methods) applied to validate significance.

The literature review, system design, and initial proposal milestones conclude with the Proposal Report Draft (August 31, 2025) and Proposal Presentation (September 8–12, 2025). Implementation and testing phases coincide with Progress Presentation I in October 2025, while integration and extended evaluation support Progress Presentation II in March 2026. The final results, reports, and viva are scheduled between April and May 2026, ensuring that the research outputs align with academic assessment deadlines.

3.7 Anticipated results

It is expected that the proposed mechanism will significantly reduce the impact of routing attacks while preserving key performance metrics. Specifically, improvements are anticipated in packet delivery ratio, network stability, and resilience to malicious behavior, with minimal additional energy or delay overhead. The integration of trust and security factors into the objective function is expected to result in more adaptive parent selection, preventing prolonged reliance on compromised nodes.

By addressing both trustworthiness and security at the routing layer, this research aims to provide a lightweight, practical, and scalable enhancement to RPL. The findings will not only demonstrate a balanced trade-off between performance and security but also contribute design insights for future IoT routing protocols.

4.0 PROJECT REQUIREMENTS

4.1 Functional Requirements (Research Implementation Requirements)

The framework should be capable of integrating trust and security mechanisms into the standard RPL protocol without introducing excessive overhead. At its core, the system should compute trust values for neighboring nodes based on metrics such as packet forwarding behavior, residual energy, and link quality, ensuring that routing decisions consider both performance and security [6], [12] [13]. The routing mechanism must also incorporate security alerts from the detection engine to mitigate threats such as sinkhole or selective forwarding attacks [2] [19] [24]. Furthermore, it is essential that the parent selection process adapts dynamically to changes in trust or network conditions, thereby ensuring stable and resilient topology formation [7]. Finally, the system must enable secure broadcasting of routing updates so that malicious manipulation of trust-related fields is prevented without incurring significant cryptographic overhead [3] [17]. These functional requirements directly support the main objective of designing a lightweight yet secure routing mechanism.

4.2 Non-Functional Requirements (Research Performance Requirements)

The system should meet several non-functional requirements to be viable in constrained IoT environments. Foremost among these is lightweight computation, ensuring that trust and security mechanisms do not overwhelm devices with limited processing power and memory [21]–[23]. The system must also demonstrate scalability, functioning efficiently across varying network sizes from small-scale IoT deployments to larger topologies [1] [24]. Reliability is another essential requirement, the system should consistently maintain acceptable packet delivery ratios and latency even under attack conditions. Similarly, resilience and adaptability are necessary, ensuring that malicious nodes can be identified and isolated while the network continues to operate without significant degradation [10] [17]. Finally, reproducibility and transparency are key research-oriented requirements, meaning that experiments and evaluation can be repeated under the same conditions by other researchers [8].

4.3 Hardware and Software Requirements

The hardware and software environment should provide a reliable platform for designing, simulating, and evaluating the proposed mechanism. Experiments will be conducted using personal computers equipped with at least 16 GB of RAM and 512 GB SSD, enabling multiple simulation instances to run in parallel for statistical accuracy. On the software side, Contiki-NG is selected as the IoT operating system, as it is widely recognized for its accurate RPL implementation and support for low-power lossy networks [8]. Its associated simulator, Cooja, provides a controlled environment to emulate realistic IoT topologies, traffic flows, and attack scenarios, making it well suited for this study. Development work will be conducted primarily in C, for protocol stack modifications, and Python, for data analysis, visualization, and statistical evaluation. Collaborative tools such as GitHub will be used for version control, while Microsoft Teams and WhatsApp will support group-level coordination.

4.4 Data Requirements

The evaluation of the proposed routing mechanism relies on both simulation-generated and real-world inspired datasets. Primary data will be collected directly from Contiki-NG/Cooja simulations, including packet traces, parent selection logs, energy counters, and routing overhead statistics. To emulate realistic attack scenarios, external datasets will also be referenced. For example, IoT-23 and UNSW-NB15 provide traffic traces containing malicious activities such as botnet traffic and intrusion attempts, which can be adapted to simulate selective forwarding or sinkhole attacks [14] [15]. Similarly, Intel Lab and SmartSantander datasets offer real-world IoT sensor traces that help design realistic traffic and energy consumption models [9]. By combining simulation data with insights from real datasets, the evaluation ensures both controlled experimental rigor and practical relevance.

5.0 BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

Item	Specification	Availability/Cost
Hardware (Personal PCs)	≥16GB RAM, ≥512GB SSD, modern CPU	Already available – LKR 0
Software (Contiki-NG, Cooja)	Open-source IoT simulation OS	Free
Libraries access (IEEE Xplore, university library)	Literature review	Free via university VPN
Version Control (GitHub)	Repository management	Free (student license)
Statistical Analysis Tools (Python, R)	Data processing and visualization	Free
Cloud resources if local capacity insufficient	50–100 hours of VM time for simulations	≈ LKR 30,000
Documentation and minor expenses	Hard-copy reports, backup tools, data transfer tools	≈ LKR 6,000

Table 1: Estimated Budget

5.2 Budget Justification

The budget for this project is minimal, primarily focused on software and cloud resources. Contiki-NG and Cooja are open-source and freely available, ensuring no cost for software tools. Personal computers with sufficient specifications (≥16 GB RAM, ≥512 GB SSD) are already available, thus eliminating the need for hardware purchases. Data access for literature review and simulation datasets is provided free via university VPN and publicly available datasets like IoT-23 and UNSW-NB15. For version control, a free student license for GitHub will be used. Statistical analysis tools such as Python and R are open-source and free. Cloud resources are only considered if local computational capacity proves insufficient, with an estimated cost of LKR 30,000 for up to 100 hours of VM time for running simulations. Miscellaneous costs, including printing, backups, and data transfer tools, are estimated at LKR 6,000. Overall, the project maintains a cost-effective approach, utilizing available resources and keeping additional expenses to a minimum.

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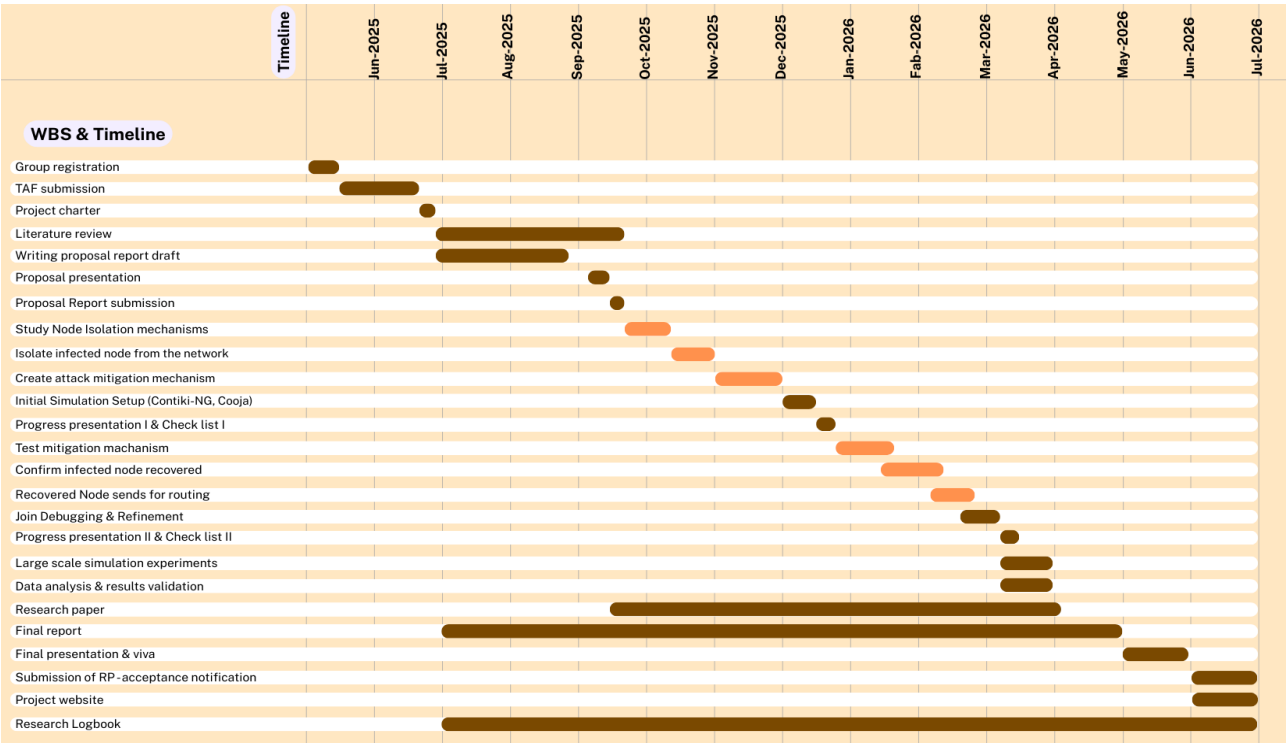
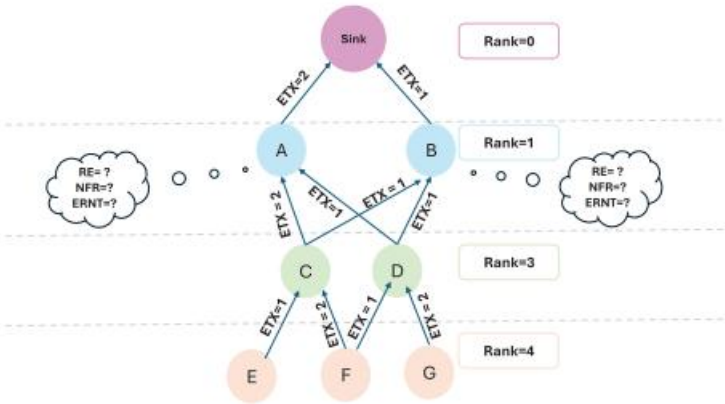
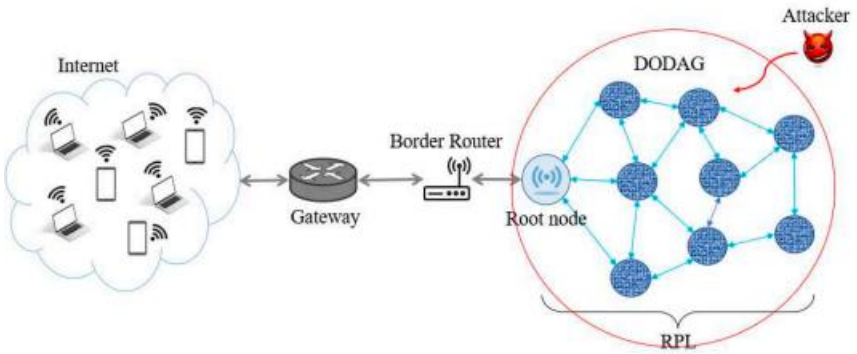
7.0 Appendices



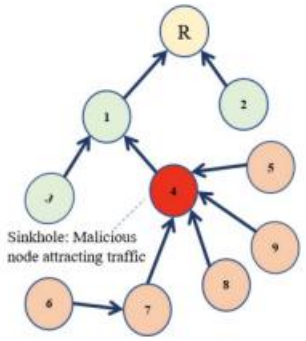
My Work Breakdown Structure

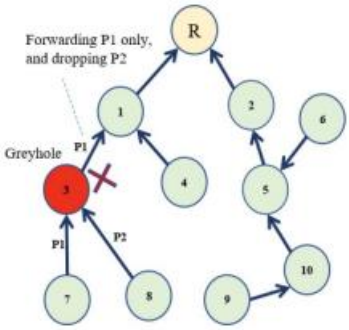


DODAG construction and parent selection

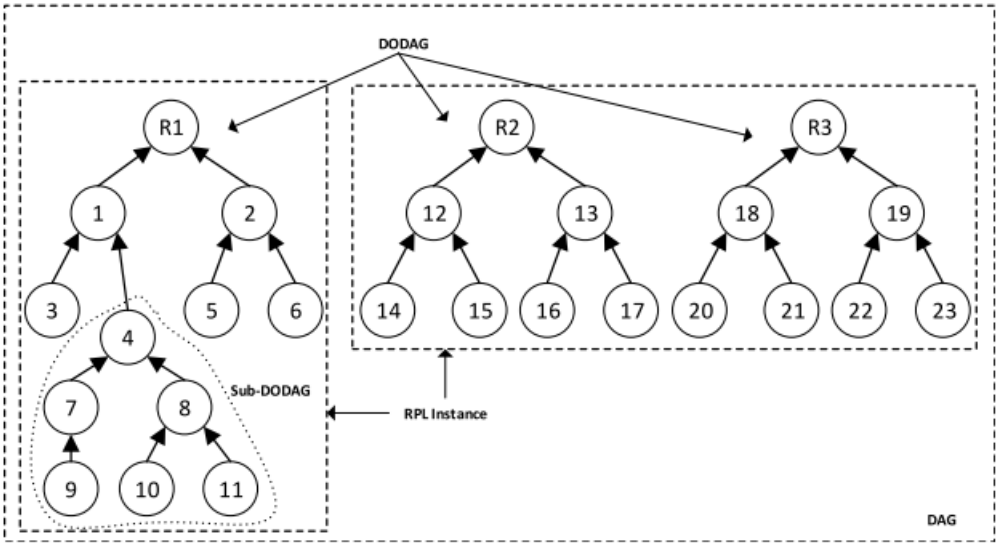


RPL-based Network in IoT

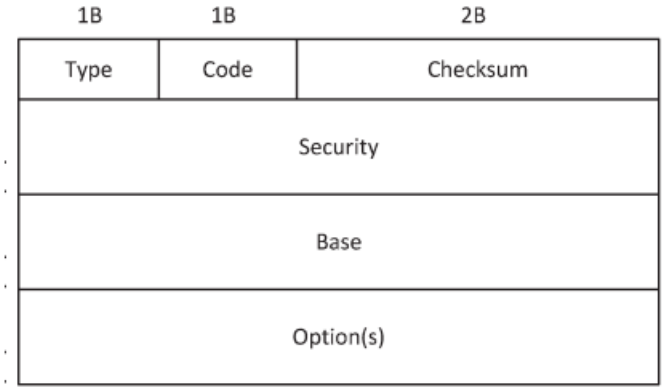




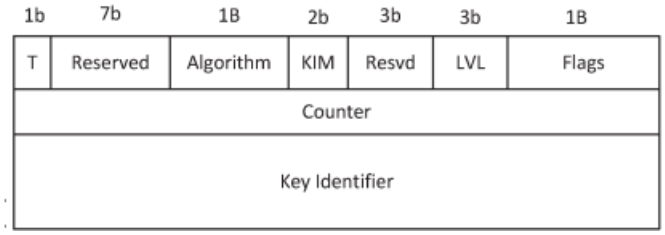
Scenario of Sinkhole attack in RPL *Scenario of Selective Forwarding attack in RPL*



An example of a typical RPL-based IoT network, showing all the building blocks: RPL Instances, DAG, DODAG, and Sub-DODAG



Secure RPL control message



Security section of a secure RPL control message.